Algorithm for AI

Assignment

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Assignment

- **Continual Concept Erasure** (**Continual CE**)
  - Suppose that three users sequentially request concept erasure as follows:
    - ✓ "**Superman** → **Van Gogh** → **Snoopy**"
  - All three concepts remain erased after the final erasure step, while all other concepts are preserved.
    - ✓ This scenario is called **Continual Concept Erasure** (**Continual CE**).

~~Superman~~

~~Van Gogh~~

~~Snoopy~~

Assignment

- **Experiments** [Individual Assignment]
  - Conduct the experiments of **Continual CE** on the following setup:
    - ✓ **Backbone**: SD v1.4
    - ✓ **Target concepts**: "Superman → Van Gogh → Snoopy"
    - ✓ **Benchmark**: Follow the setting from [1].
  - Submit the **code** and **report**.
    - ✓ Please describe your **problem** (**motivation**), **idea**, and **experimental results** in the report.
      - Up to 4 pages (including **algorithm**, quantitative and qualitative results, ablations, and analysis).
      - **Format**: You can download the latex & word templates from the **CVPR** homepage.
      - **Due**: ~ 4/21 (to the LMS site).

[1] M. Lyu *et al.*, "One-dimensional Adapter to Rule Them All: Concepts, Diffusion Models and Erasing Applications", *CVPR 2024*.

Assignment

- **Experiments** [Individual Assignment]
  - Example of Quantitative Table:

Concept	**Superman$_e$**	**Superman$_r$**	**Van Gogh$_e$**	**Van Gogh$_r$**	**Snoopy$_e$**	**Snoopy$_r$**	**MS-COCO**	
	CS ↓	FID ↓	CS ↓	FID ↓	CS ↓	FID ↓	CS ↑	FID ↓
SD v1.4	(fill in the measured values using the original SD v1.4)							
UCE (baseline) …								
Ours								

※ Superman$_r$ = {Batman, Thor, Wonder Woman, Shazam}

※ Van Gogh$_r$ = {Picasso, Monet, Paul Gauguin, Caravaggio}

※ Snoopy$_r$ = {Mickey, Spongebob, Pikachu, Hello Kitty}

Assignment

- **Presentation** [Team Assignment]
  - Present **problems** and their solutions (**ideas**) proposed by team members.
  - Show **experimental results** with the **ablation study**, and compare with individual models.
    - ✓ Check the synergetic performance of the team's model on the same benchmark.
  - Include **analysis** of why the team's model shows lower/better performance than individual models.

  - Please submit **PPT** (due: **4/21**) to LMS.
    - ✓ Team order will be announced on **April 22th** (the date of presentation).
    - ✓ **Duration**: 20 minutes (presentation) + 10 minutes (Q&A) for each team (total 30 min.).
    - ✓ No restrictions on the slide format.
    - ✓ *The presentation should be conducted in English.*